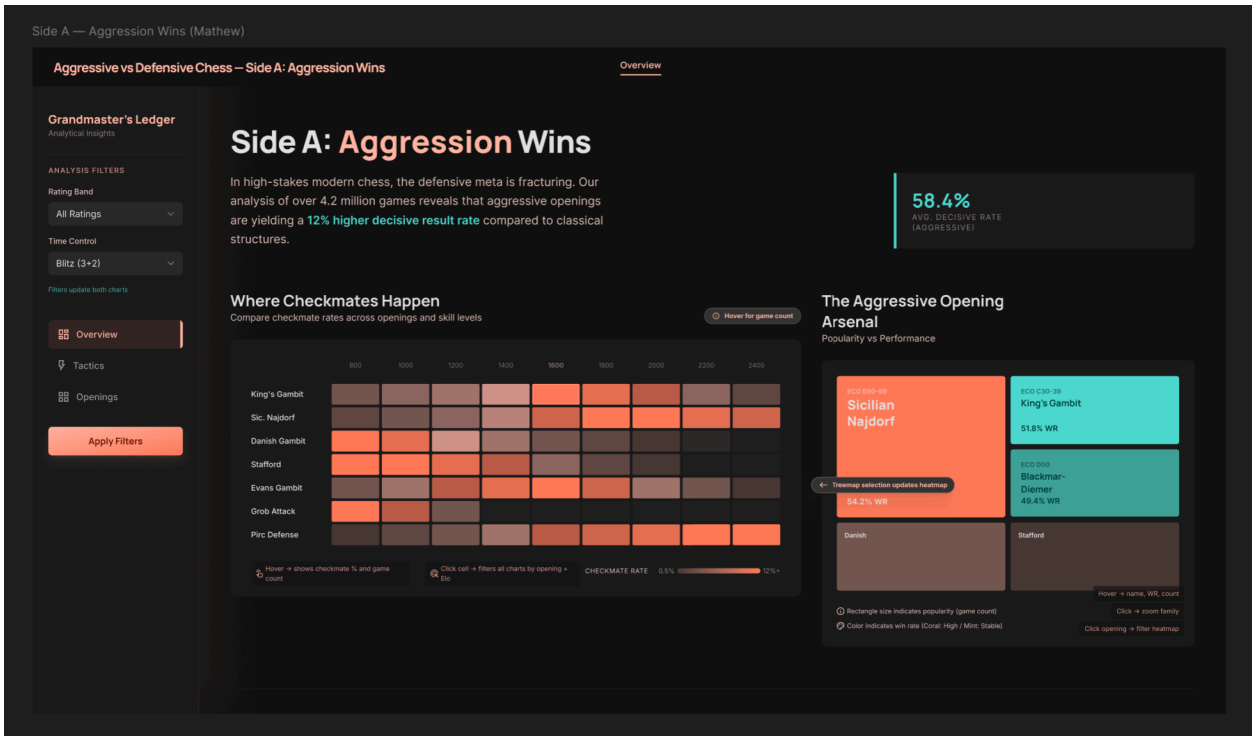
CSE 578 Course Project

Project Sketches

Team Members:
1) Mathew Abraham Palliambil - mpalliam@asu.edu
2) Basil Shibu Thomas - bthoma70@asu.edu
3) Asmit Datta - adatta18@asu.edu
4) Chandana Priya Srinivasa - csrini10@asu.edu
5) Kruthika Kadurhalli Raghu -kkadurha@asu.edu
6) Parth Jain - pjain121@asu.edu

Perspective A: Aggressive wins



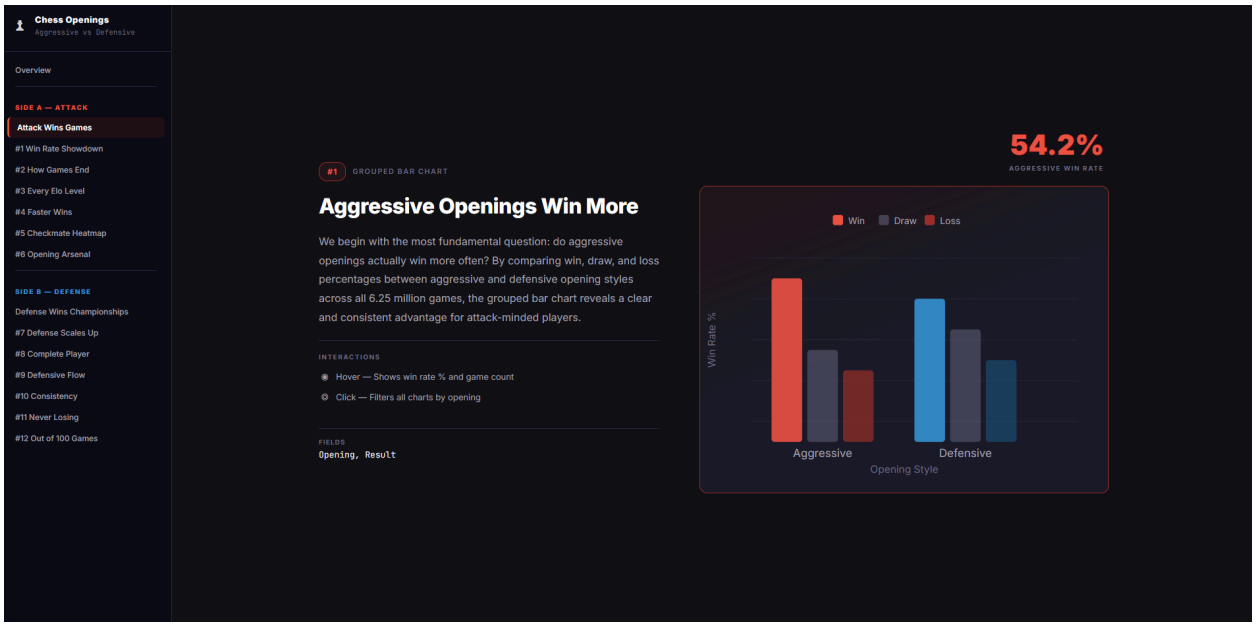
This sketch depicts the Side A interface of our dual visual data narrative, which presents the case that aggressive chess openings result in more decisive and successful outcomes in practice. The objective of this side of the narrative is not only to present data, but to take the reader through a compelling narrative of the process of how aggressive chess opening strategies result in advantageous outcomes in actual games.

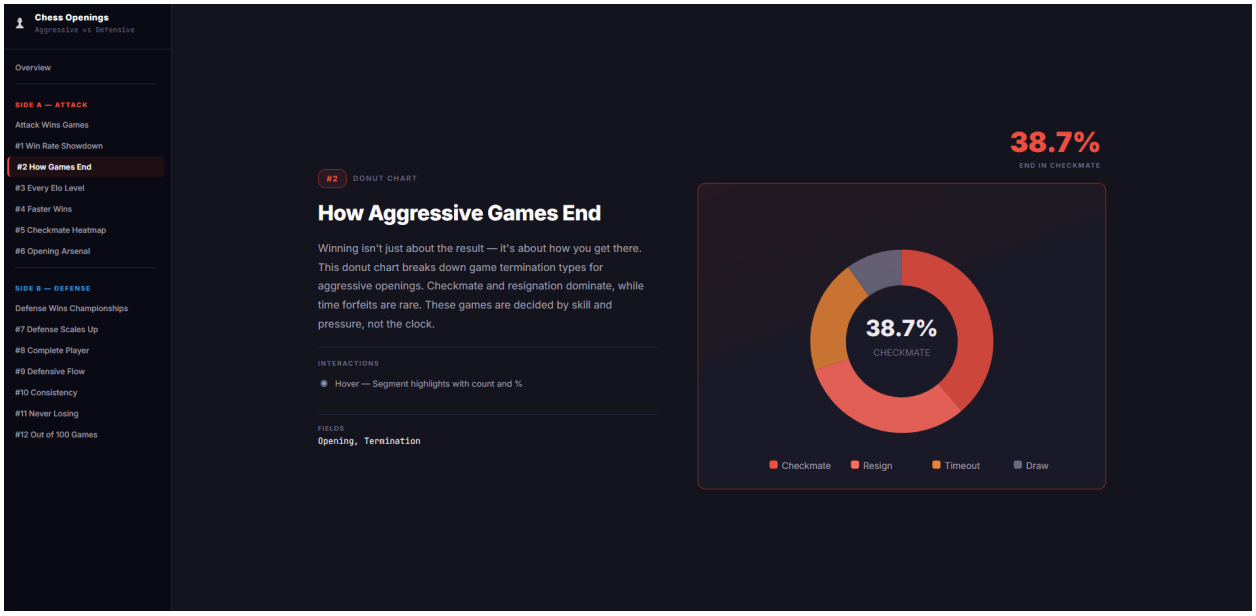
The narrative begins with users exploring the Opening Arsenal treemap, which presents an overview of aggressive chess opening strategies based on popularity and win rates. This enables users to see which opening strategies are popular and effective. After selecting an opening, users are presented with a deeper exploration of the opening strategy through the heatmap, where users can explore checkmate rates based on Elo ratings. This presents users with the understanding that aggressive chess opening strategies result in decisive outcomes regardless of the user's skill levels.

Through the interactive process, users are able to navigate from a general understanding of chess opening strategies to a deeper understanding of the process of how aggressive chess opening strategies result in advantageous outcomes, thus supporting the narrative of the effectiveness of aggressive chess opening strategies.

- Users can start by exploring the treemap and finding popular and successful aggressive openings

- Selecting an opening updates the heatmap, allowing users to compare the performance of the selected opening at various Elo ratings
- The heatmap also shows the high checkmate rate of aggressive openings across skill types
- Hovering over the heatmap gives users detailed values, such as the checkmate rate, win rate, and number of games, to help interpret the results
- Global filters, such as the rating band and time control, update all views to reflect the impact of the selected filters on the performance of the opening
- Coordinated views help users navigate from exploring the treemap to comparing the performance and finally understanding the reason behind the success of the aggressive openings.





Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#3 LINE CHART

Aggression Works at Every Level

A common counterargument is that aggression only works against weak players. This line chart tracks aggressive opening win percentages across Elo brackets from 800 to 2400. The advantage holds at every skill level — attack works whether you're a beginner or approaching grandmaster territory.

INTERACTIONS

- Hover — Shows win rate % and game count
- Click — Filters charts by opening + Elo
- Double click — Reset Elo distribution view

FIELDS

`Opening, WhiteElo, Result`

84.2%

MAX EDGE POTENTIAL

Chess Openings
Aggressive vs Defensive

Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#4 SCATTER PLOT

Aggressive = Faster Wins

Time is a resource in chess. This scatter plot maps each aggressive opening by its average game length versus its win percentage. The data clusters in the high-win, short-game quadrant — aggressive openings don't just win more, they win faster.

INTERACTIONS

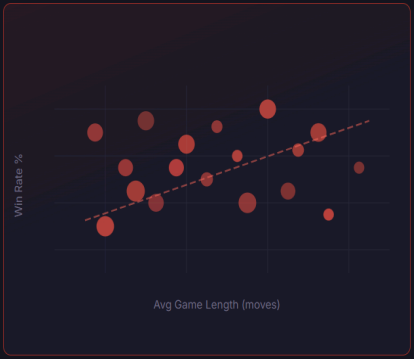
- Hover — Opening name + detailed stats
- Brush — Drag to filter by range

FIELDS

`Opening, AN (move count), Result`

26

AVG MOVES TO WIN



Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#5 HEATMAP

Where Checkmates Happen

This heatmap reveals checkmate rates across the top 10 aggressive openings and Elo brackets. Each cell's color intensity represents how often that opening produces a checkmate. The pattern is striking: aggressive openings generate checkmates consistently across all levels of play.

INTERACTIONS

- Hover — Cell value and checkmate count

FIELDS

`Opening, Termination, WhiteElo`

10
OPENINGS ANALYZED

Chess Openings
Aggressive vs Defensive

Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#6 TREEMAP

The Aggressive Opening Arsenal

Finally, we explore the full landscape of aggressive openings. This treemap shows every aggressive opening family, sized by game count and colored by win rate. Click into any family to reveal its variations — a vast and diverse arsenal of weapons.

INTERACTIONS

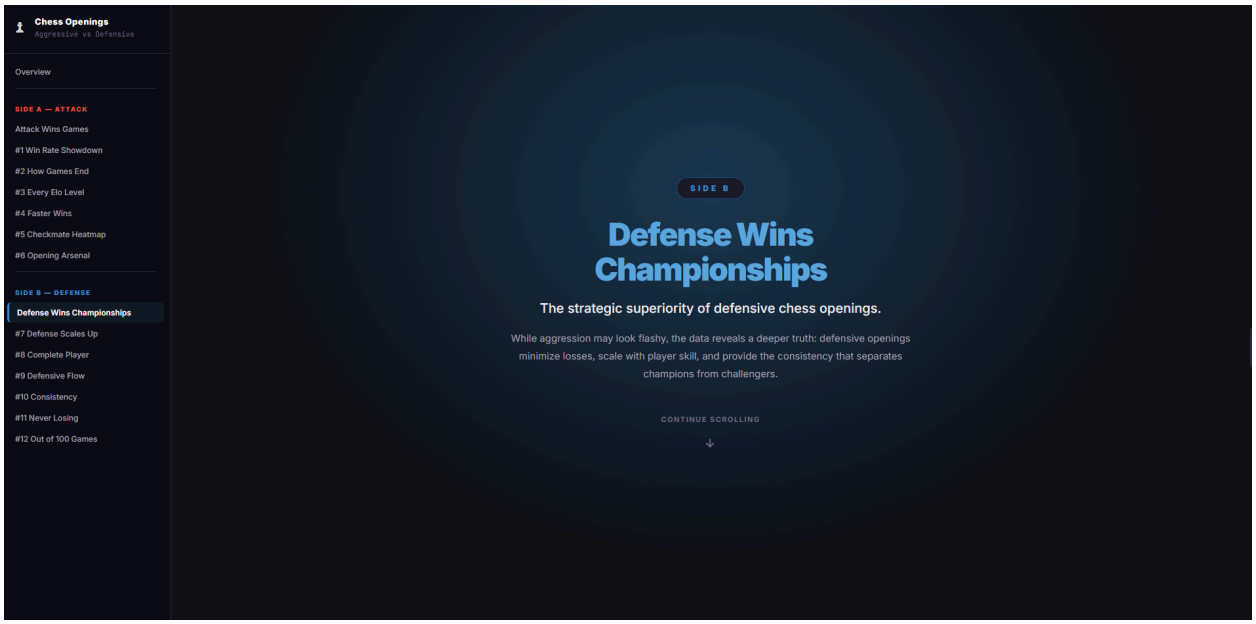
- Click — Zoom into opening family
- Hover — Win rate and game count

FIELDS

`Opening, ECO, Result`

47+
AGGRESSIVE VARIATIONS

Perspective B: Aggressive vs Defensive Play: Risk vs Consistency



This series of visualizations will demonstrate the narrative of Side B of our project, highlighting how defensive chess openings contribute to a stable and consistent game. The narrative of Side B of the project will be told by a series of visualizations. The first is a stacked area chart that demonstrates how, as the rating of a player increases, there is a significant trend towards defensive openings such as the Caro-Kann, French, and Berlin Defenses. This shows how stronger opponents are likely to play a safer game. The radar chart explains why there is a trend towards defensive openings by showing a comparison between defensive and aggressive play in various performance metrics. Defensive play is shown as a more balanced and even distribution, showing a stronger and more consistent game, whereas aggressive play is shown as a less even distribution, showing a less consistent game with many weaknesses. From the Sankey diagram above, we see how defensive openings lead more often to a stable game where a number of games result in a draw. On the other hand, aggressive openings lead to a less stable game where a number of games result in a win or a loss. The box plot shows how defensive is a more consistent game with less variance, showing less.

This is done in a way such that the entire system is designed with coordinated interactions, such as hover interactions where the user can see detailed values such as percentages and counts, click and toggle interactions where the user can filter between aggressive and defensive styles, and global filters such as rating band and time control, where all the views are updated at once. Scrollytelling is also used,

where the user is taken through the narrative, and each chart is loaded progressively as the user scrolls, with animations and annotations to highlight the insights. This is done in a cohesive way such that the user not only sees what happens in defensive play but is also made to understand the reasons behind it leading to more reliable and controlled outcomes than the use of aggressive strategies.

Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#7 STACKED AREA CHART

Defense Gets More Popular as Players Improve

This interface highlights defensive openings across rating levels, showing that higher-skilled players prefer more stable strategies like the Berlin defense. It includes hover details and filters for rating and time control, enabling easy exploration and comparison of trends.

INTERACTIONS

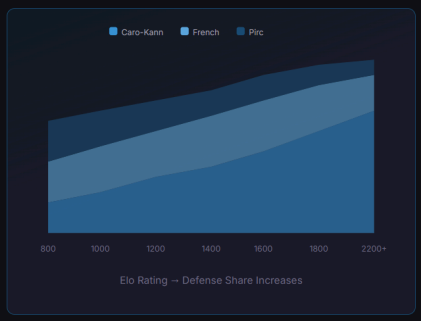
- Hover — Opening name + % share
- Scroll — Triggers Elo transition animation

FIELDS

`Opening, WhiteElo, BlackElo`

67.3%

DEFENSE AT 2200+ ELO



Chess Openings
Aggressive vs Defensive

Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#8 RADAR / SPIDER CHART ★

The Complete Player Chooses Defense

This interface shows that defensive openings become more popular at higher rating levels and lead to more stable outcomes. The stacked chart highlights increased usage of openings like the Caro-Kann, French, and Berlin, while the radar chart shows that defensive play is more balanced and consistent than aggressive play.

INTERACTIONS

- Hover — Exact values per axis
- Toggle — Individual openings on/off
- Scroll — Animated morph transition

FIELDS

`Opening, Result, Termination, AN, WhiteRatingDiff`

6

DIMENSIONS COMPARED

#11 DIVERGING BAR CHART

Defense Means Never Losing

Winning matters, but not losing matters more. This diverging bar chart centers at zero — bars extend left for losses and right for wins. Defensive openings have dramatically shorter loss bars. You may not always win, but you almost never lose badly.

INTERACTIONS

- Hover — Exact win/loss percentages
- Sort — Toggle by win rate or loss rate

FIELDS
`Opening, Result`

8.4%
AVG LOSS RATE

Defensive Openings: Win vs Loss Rate

Chess Openings
Aggressive vs Defensive

Overview

SIDE A — ATTACK
Attack Wins Games
#1 Win Rate Showdown
#2 How Games End
#3 Every Elo Level
#4 Faster Wins
#5 Checkmate Heatmap
#6 Opening Arsenal

SIDE B — DEFENSE
Defense Wins Championships
#7 Defense Scales Up
#8 Complete Player
#9 Defensive Flow
#10 Consistency
#11 Never Losing
#12 Out of 100 Games

#12 WAFFLE CHART

Out of 100 Games, How Many Do You Lose?

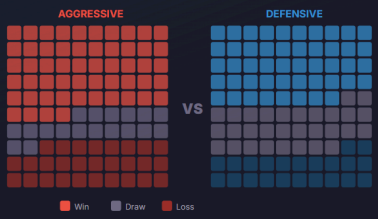
Sometimes the simplest visualization tells the most powerful story. Two side-by-side waffle charts show win, draw, and loss proportions per 100 games. Count the red squares. The visual difference is stark: defense doesn't just reduce losses — it nearly eliminates them.

INTERACTIONS

- Hover — Square highlights with tooltip
- Scroll — Animated fill on scroll

FIELDS
`Opening, Result`

100
GAMES PER GRID



Each square = 1 game out of 100